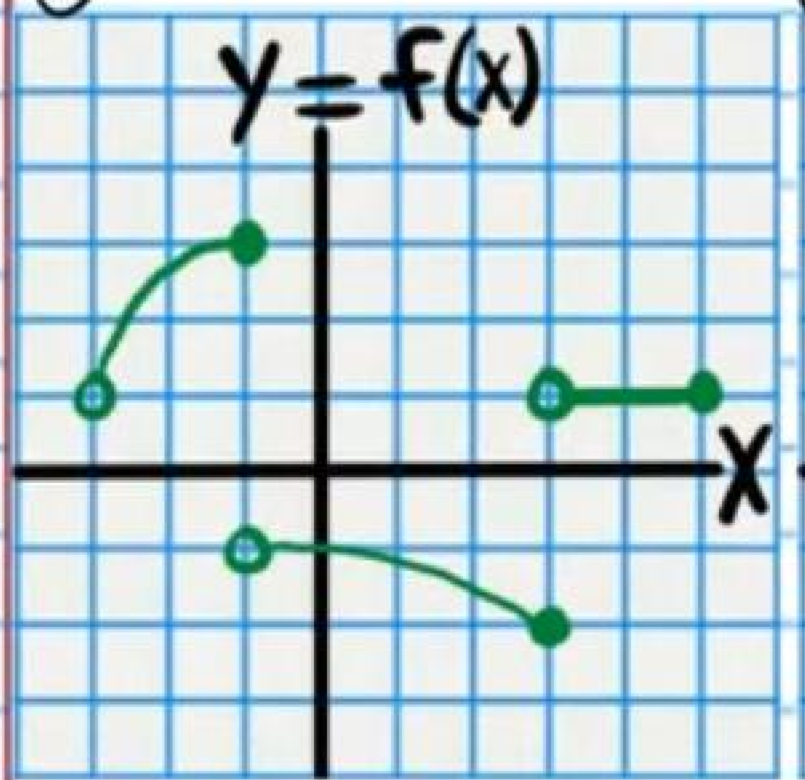
Name
Date
Course

Applications of Differential Calculus (Maximum and Minimum Values) Homework

Write the x-value where each maximum or minimum occurs.

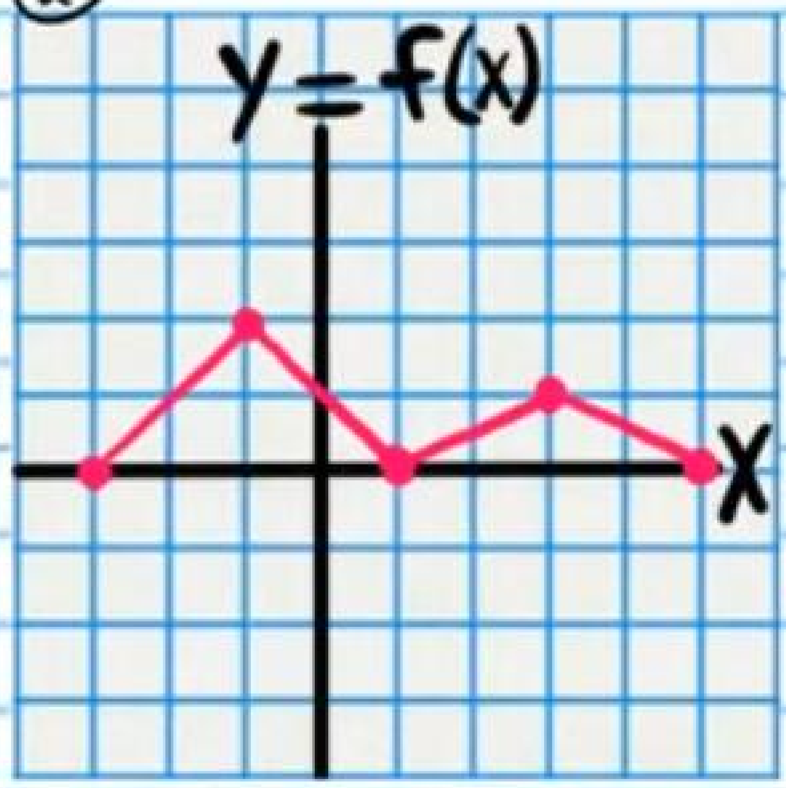
* Assume that the direction of each graph continues as indicated by the arrows outside the window.

①



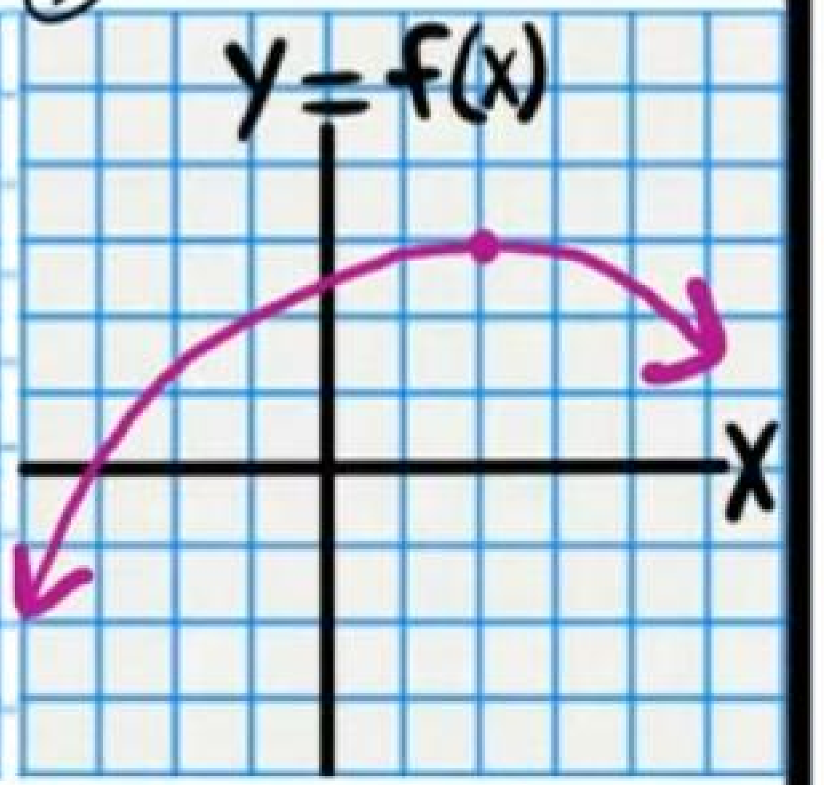
Absolute Max at
Absolute Min at
Local Max(s) at
Local Min(s) at

②



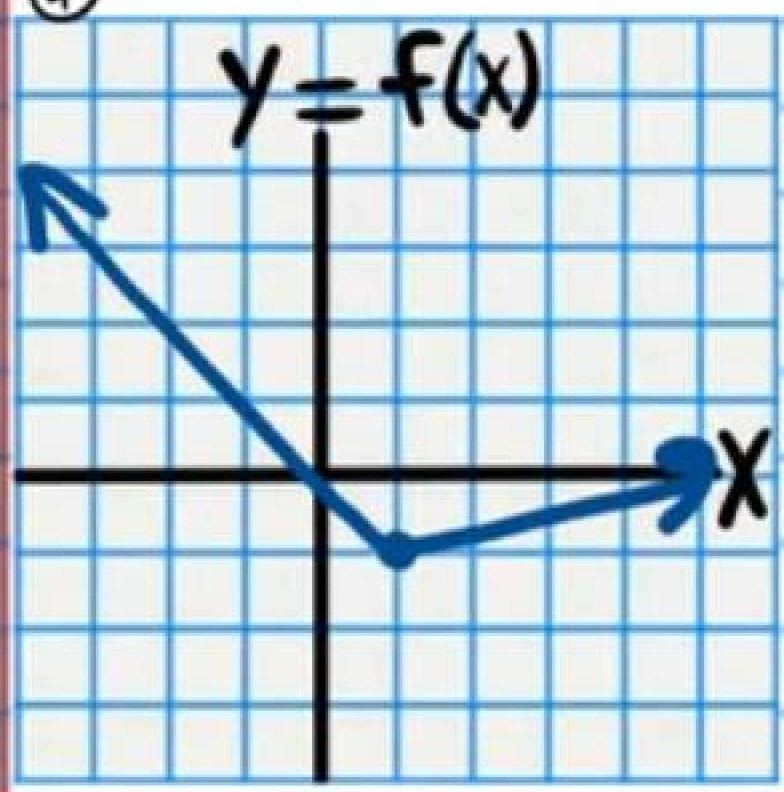
Absolute Max at
Absolute Min at
Local Max(s) at
Local Min(s) at

③



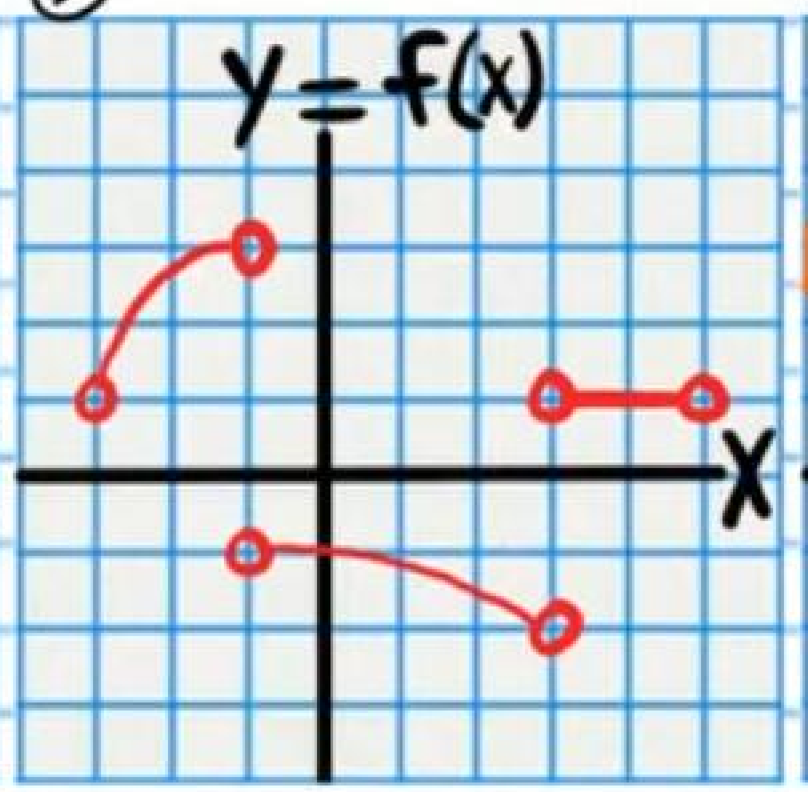
Absolute Max at
Absolute Min at
Local Max(s) at
Local Min(s) at

④



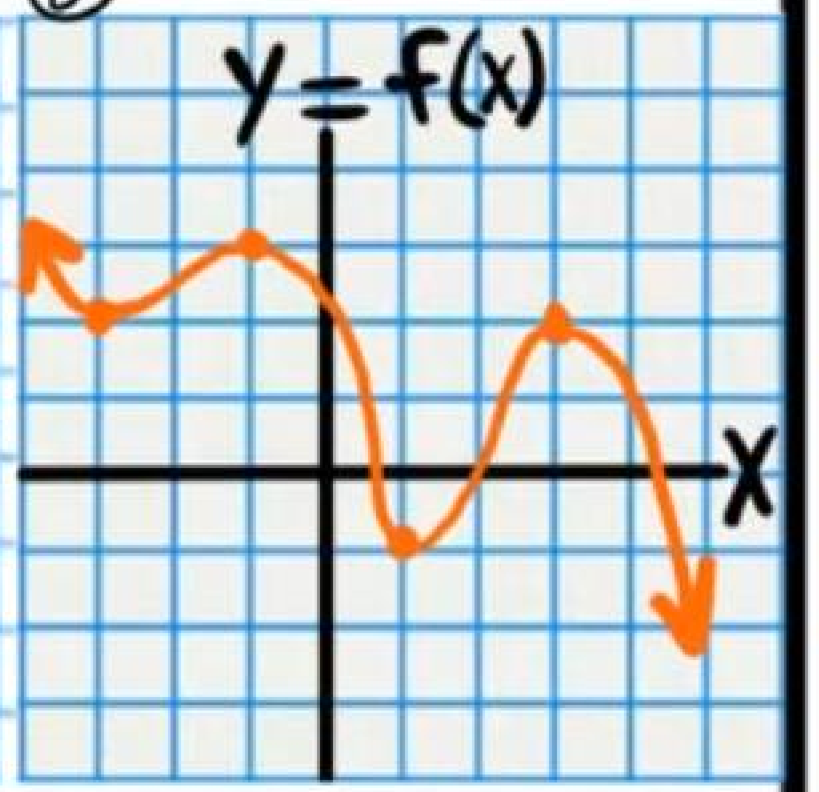
Absolute Max at
Absolute Min at
Local Max(s) at
Local Min(s) at

⑤



Absolute Max at
Absolute Min at
Local Max(s) at
Local Min(s) at

⑥



Absolute Max at
Absolute Min at
Local Max(s) at
Local Min(s) at

Find the absolute maximum and the absolute minimum of each function on the specified interval.

★ A graphing calculator will help us with calculations and concepts.

(7) $f(x) = \frac{3}{5}x^5 - \frac{1}{2}x^4 - \frac{1}{3}x^3$, $[-\frac{1}{2}, 2]$

I) Finding Critical Numbers

II) Evaluating the Function at the Critical Numbers

III) Evaluating the Function at the Endpoints

Absolute Maximum:

Absolute Minimum:

★ You must use the chain rule when finding the derivative.

⑧ $f(x) = -2\sqrt[3]{x^2-1}$, $[-2,2]$

I) Finding Critical Numbers

II) Evaluating the Function at the Critical Numbers

III) Evaluating the Function at the Endpoints

Absolute Maximum:

Absolute Minimum:

⑨ $f(x) = \sqrt[3]{x^2} - x$, $[-1, 1]$

I) Finding Critical Numbers

II) Evaluating the Function at the Critical Numbers

III) Evaluating the Function at the Endpoints

Absolute Maximum:

Absolute Minimum:

⑩ $f(x) = 2\sqrt[5]{(x-2)^4} - (x-2)^2 - 2, \quad [1,3]$

I) Finding Critical Numbers

II) Evaluating the Function at the Critical Numbers

III) Evaluating the Function at the Endpoints

Absolute Maximum:

Absolute Minimum:

Homework Answers

①

Absolute Max at $x=-1$

Absolute Min at $x=3$

Local Max(s) at $\emptyset$

Local Min(s) at $\emptyset$

②

Absolute Max at $x=-1$

Absolute Min at $x=-3,1,5$

Local Max(s) at $x=-1,3$

Local Min(s) at $x=1$

③

Absolute Max at $x=2$

Absolute Min at $\emptyset$

Local Max(s) at $x=2$

Local Min(s) at $\emptyset$

④

Absolute Max at $\emptyset$

Absolute Min at $x=1$

Local Max(s) at $\emptyset$

Local Min(s) at $x=1$

⑤

Absolute Max at $\emptyset$

Absolute Min at $\emptyset$

Local Max(s) at $\emptyset$

Local Min(s) at $\emptyset$

⑥

Absolute Max at $\emptyset$

Absolute Min at $\emptyset$

Local Max(s) at $x=-1,3$

Local Min(s) at $x=1,-3$

⑦

Absolute Maximum: $\boxed{8.5\overline{3}}$ Absolute Minimum: $\boxed{-\frac{7}{30}}$

⑧

Absolute Maximum: $\boxed{2}$ Absolute Minimum: $\boxed{-2.884499}$

⑨

Absolute Maximum: $\boxed{2}$ Absolute Minimum: $\boxed{0}$

⑩

Absolute Maximum: $\boxed{-0.96587}$ Absolute Minimum: $\boxed{-2}$